Learning Objective 7.3: I can use Ampere's Law to calculate the magnetic field at an observation point for simple magnetostatic systems.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **A coax with a thick shield.** A coaxial line has an inner conductor of radius $a = 1$ mm and a shield that fills $b = 4$ mm $< \rho < c = 5$ mm. It carries $I = 2$ A at DC, out along the inner conductor and back along the shield, with the current uniform over each conductor's cross section.

   (a) Use Ampere's law to find H_ϕ in all four regions.

   (b) Find H_ϕ at $\rho = 0.5$ mm, 2 mm, 4.5 mm, and 6 mm.

   0.5 mm: ____

   2 mm: ____

   4.5 mm: ____

   6 mm: ____

2. **A finite solenoid.** A coil of $N = 1000$ turns is wound over $\ell = 20$ cm on a form of radius $a = 2$ cm and carries $I = 0.5$ A, in air.

   (a) Find H and B inside, treating the coil as a long solenoid.

   H =

   B =

   (b) Use the finite-solenoid formula to find B at the center and at one end of the coil.

   center:

   end:

   (c) How long must a coil of radius 2 cm be for its center field to come within 1% of nI?

   ℓ =

3. **A toroid.** A toroid of $N = 400$ turns has a rectangular cross section with inner radius $r_1 = 2$ cm, outer radius $r_2 = 3$ cm, and height $h = 1$ cm. It carries $I = 0.25$ A and has an air core.

   (a) Find H at the inner and outer edges of the core.

   $H(r_1)$ = ☐

   $H(r_2)$ = ☐

   (b) Derive the magnetic flux Φ through one turn, the rectangle $r_1 < \rho < r_2, 0 < z < h$, and evaluate it.

   (c) The current path passes through the core N times, so the flux it links is $N\Phi$. Find the inductance $L = N\Phi/I$.

   L = ☐

4. **RG-59.** RG-59 has $a = 0.29$ mm, $b = 1.85$ mm, and solid polyethylene with $\epsilon_r = 2.25$. Lesson 25's practice set found $C' = 67.5$ pF/m.

   (a) Find L', Z_0, and v_p.

   L' = ☐

   Z_0 = ☐

   v_p = ☐

   (b) What ratio b/a gives $Z_0 = 50\,\Omega$ with the same dielectric?

   b/a = ☐

Documentation: _____